

Tommy (Zuyu) Mao

Network Automation & Infrastructure Engineer

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Experience

Ericsson · Network Automation & Infrastructure Engineer

Feb 2024 - Present

Telstra Network CI/CD

- Owned automated release validation and regression pipelines for **10,000+ distributed production nodes**, ensuring safe rollouts, high availability, and rapid rollback.
- L2 on-call incident response, mitigation, service restoration, and RCA-driven remediation across production network incidents.
- Built a scalable Python-based automation framework (Python, FastAPI, Bash, YAML, REST APIs, SQLite), reducing regression cycles by **50% (2 days → 1 day)**, now used as the daily CI/CD backbone.
- Designed upgrade-safe network features that preserved critical service continuity during nationwide rollouts; led lab operations spanning Layers 1-7 to validate real-world failure modes and traffic flows.

NBN Network Integration

- Primary engineer responsible for development, operation, and reliability of core network automation tooling.
- Configured and deployed transport and backhaul infrastructure across **3,000 geographically distributed rural sites**.
- Built an automated licensing pipeline, reducing provisioning turnaround from **3 days → 1 day** (Python, Playwright, REST APIs).

Optus · Solutions Consultant

Dec 2022 - Feb 2024

- Ranked **#1 sales consultant** for three consecutive months, consistently exceeded both sales and customer-outcome KPIs.
- Consulted residential and small-business customers across mobile, broadband, NBN, 4G/5G, and WiFi; delivered technical recommendations clearly in English and Mandarin.

Selected Projects

Release Validation Pipeline · custom Ansible-style CI framework

Python, FastAPI, YAML, SQLite, GitLab CI, ADB

- Built from scratch: YAML playbooks describe router/hardware config and commands; Python executes against real hardware on internal Linux servers; ADB drives OS pushes; pass/fail validators auto-populate SQLite and a live dashboard. Cut deployment window from 13h to 9h per upgrade.

LLM Wiki - AI Knowledge Base · multi-agent ingestion pipeline

Claude API, Local LLaMA, Python, FastAPI, SQLite

- 9 sources (Gmail, Slack, Drive, Obsidian, Notion, plus Telegram/WhatsApp via OpenClaw). Local LLaMA handles bulk extraction; Claude Haiku cleans edge cases; Claude Sonnet writes structured wiki pages. 13,643 items ingested, 53 searchable pages.

Self-Hosted Home Lab · full stack on a single Linux server

Docker, Linux, Unifi, Home Assistant, Cloudflare ZT, AWS S3

- Ubuntu 24.04 host running 15 containers; Unifi Dream Machine routing with VLAN-isolated IoT subnet; Zigbee2MQTT → Home Assistant automations; three Cloudflare Zero Trust tunnels (zero open inbound ports); AWS S3 nightly backup with failover.

Skills

Network Automation: Python, FastAPI, Bash, YAML, REST APIs, GitLab CI/CD, custom Ansible-style frameworks

Networking: BGP, OSPF, IS-IS, MPLS, IPsec, TCP/IP, DHCP, DNS, QoS, SNMP, Leaf-Spine, VPNs, HTTP/S, SSL/TLS

Systems & Cloud: Linux, Docker, FastAPI, SQLite, nginx, AWS EC2, AWS S3, Cloudflare Zero Trust

Platforms: Cisco IOS XR/XRd, Nokia SR Linux, Arista EOS, Palo Alto PAN-OS, Ericsson RAN/Microwave/Radio/Routers, Unifi

AI / LLM: Claude API, ChatGPT API, Local LLaMA, RAG, multi-agent, tool-use, Playwright

Languages: Fluent in English and Mandarin

Education & Certifications

Bachelor of Science · Electrical Systems

University of Melbourne

Cisco Certified Network Associate (CCNA) · studying CCNP